

Notes for a Course on Proofs

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To the Instructor

The following notes for an “introduction to proofs” course have developed over the last several years, and have been gleamed from many sources and edited to my purposes and for my students at Sam Houston State University. You will want to do the same for your students - edit the problems based on their ability and the needs of your course. You may choose to delve more in-depth into some topics or to skip some problems entirely in favor of covering another topic. I want to spend the first part of this introduction discussing the philosophy of the notes as they have developed, and will then provide more detailed information on how I have used the notes, how I have decided to assess students working through these notes, and the specifics of how my version of this class runs.

While my course is a modified Moore method course, during the first class meeting I spend very little time talking about the syllabus, the course format, or the philosophy of the Moore method. Instead I distribute some subset of the first nine problems (in Chapter 1). Students are allowed to work in groups to discuss these problems. This, as a matter of fact, is the only time that they are allowed to discuss problems with anyone except the instructor. They are told that they can solve these problems by “any means necessary” for the next class period. We then begin presentations on the second class day. I love all nine of these problems, and so choosing a subset to distribute on the first day is difficult. I try to hold some of these problems back in case I teach this course two semesters in a row, or two years in a row. Since the students are allowed to use “any means necessary” on these problems (which means that they might talk to other students about these problems) I want to make sure that they are not solving the problems too easily by finding one person from the previous semester who has seen all of these problems and can provide solutions to all of them.

When students present the second day, I ask “How did you figure this out?” It makes my day when they say “I Googled it” or “I was talking to some friends about these problems” or something similar. Then I have the chance to say “That was a great idea. You can never do that again with any problems in this class.” It brings home to them that when I say that they cannot speak to anyone else about the problems, I mean not just that they cannot speak to classmates, but to anyone. When I say that they cannot use

any outside resources, I mean *no* outside resources - no internet, no books, no Google. It also gives me a chance to get some students to the board who feel confident in their solutions (because they looked them up). However, there are also problems on the list that they cannot find online (or at least they couldn't be found easily the last time I checked).

These problems also show students that they will have to think about things differently than they have in other mathematics courses - that the course will be challenging, require critical thinking, and that they will have to work hard all semester.

On the second day, students are given the first set of "real" problems - starting with some basic definitions and number theory proofs to get them started. There are usually some students who are ready to present these by the third class period and then we are off and going. I give a lot of homework problems at the beginning of the semester until students realize that they are not expected to complete *all* of the problems before the next class period. Sometimes this happens quickly. Sometimes not. But I want to make sure that they *always* have something to work on.

That brings up another point - I do not ever distribute all of the problems at once. I dole out the problems a page or so at a time, so that I have some control over which problems will be presented. After all, I don't want to be talking about basic logic and have a student decide that they want to present problems about induction. Therefore, the students get 15 or so problems when they seem to be ready for them.

0.1 The Instructor's Role in the Classroom

I want to emphasize that the role of the instructor changes over the course of the semester. In the beginning, my role is to get students to the board, help them feel comfortable presenting, and to help the students in the room feel comfortable asking questions. Frequently in the beginning, I will rephrase questions posed by students in the class, so that they learn to ask questions more effectively (and following the guidelines that I mention below). Later in the semester, my role is to moderate the discussion (deciding whose turn it is to speak next) and to decide if the presenter needs to have a chance to re-evaluate their solution and try again the next day. This last is saved for a situation when several students in the class seem to have spotted a potentially fatal flaw in the solution being presented. At that time, I will give the presenter a chance to think for a moment or two, but, if no fix for the proposed flaw is forthcoming, will ask the student if they want to try the problem again at the beginning of the next class period.

It is also, always, my job keep track of which problems have been presented. I've run the course in different ways - by allowing students who

have a solution to the next problem in the series to volunteer or by calling on a student at random and allowing them to choose any open problem to present being two of my favorites. In recent years, the latter has been my method of choice. I have designed the notes so that everyone should have something that they have solved at each class period. In order to do this, the problems range from examples helping students practice with definitions to problems which are, some semesters, never solved.

I try to remember that my role in the classroom is to moderate discussion and not to *lead* discussion. I moderate the discussion by calling on students, making sure that the presenter understands the question, and making sure that the questioner understands the answer. I pause to make sure that the students in the class agree that the solution is complete. (By the way, when they think that the solution is complete and correct, but it's not, the problem appears on an exam or quiz.) Early in the semester I might stop the class and say "Someone should have a question now," and then hold the class until they figure out what the error is. Sometimes I'll ask the presenter, "Are you sure?" or "How do you get from line 3 to line 4?" even when there isn't a problem.

One thing that you realize very quickly in this type of course is that the presenter always watches the instructor to see if the solution is correct. So does the rest of the class, by the way. Everyone watches to see if you, the instructor, flinch during the presentation. I try not to show my hand and look as confused when it is a correct presentation as when there is an error; or, perhaps, I should say that I look equally pensive whether the solution is correct or not.

So, all in all, the instructor is responsible for maintaining the tone of the classroom and for keeping everyone up to date about which problems have been presented and which are still open. The instructor is also responsible for grading - in this course, this is an extremely important aspect as the goal of the course is get students to be able to communicate mathematics. There are several sections below about how I have chosen to grade my course, and I include that discussion as a starting point for your own planning.

0.2 The Course Content

As mentioned above, these notes are developed for and have been used in a semester-long "Introduction to Proofs" course. They have been used and have evolved over several years at Sam Houston State University. The official course description is brief: "This course includes an introduction to sets, logic, the axiomatic method and proof." Many of the specific topics to be covered are left to the instructor, but it is traditional to cover the following:

- Logic, including inverse, converse, contrapositive and negation

- Techniques of proof, including proof by contradiction, proof by induction, and direct proof
- An introduction to number theory
- Set theory (focusing on definitions in set theory and set operations), including families of sets
- Relations and, sometimes, functions
- Partitions and equivalence classes

We do want to ensure that students who complete the course have an understanding of the following proof techniques:

- Direct proof
- Proof by contradiction
- Proof by contrapositive
- Induction

There is some freedom given to the instructor to choose the topics and depth of material in those topics to ensure that students have some exposure to sets, relations, and partitions, while still mastering proof techniques. I find that I cover number theory, set theory, and relations well, while not worrying about completing a discussion of functions, partitions, and equivalence relations. By the time successful students complete my course, they are comfortable with reading a definition, developing examples and non-examples, and proving new facts from these definitions.

I want to point out that there is no section specifically devoted to the study of quantifiers, but these concepts do appear in the notes through the problems presented. I make sure that I discuss these issues (eg., how do you negate a universal quantifier) at the appropriate times. An instructor using these notes should feel free to add specific problems to address these issues during the chapter on logic, if that would be beneficial to your students.

0.3 Information about Sam Houston State University and Our Students

Sam Houston State University is a medium-sized state school. There are about 16,000 undergraduates on campus, and we have about 90 mathematics majors, many of whom are minoring in secondary education with the hopes of teaching high school. Many of our students are first-generation college students, and many of them did very well in high school. Many of them

do not, initially, understand the difference between high school courses and college courses. Luckily, they take the “Introduction to Proofs” course in their first semester of their sophomore year (or, very occasionally, in the second semester of their freshman year). This means that they have some other mathematics at the college level already under their belt, and they understand that they will have to work differently (i.e., harder) than they did in high school. However, this is also their first non-calculus class in the major, and they are surprised by the change. Again, luckily, they have been warned by upper classmen that this is a challenging course, and that they will need to focus much of their attention on the material presented here.

This course usually runs with approximately 25 students, although I’ve had closer to 30 or more some semesters. The grading compromises that I mention below came about because of this relatively large class size. I found it was impossible to grade every problem I assigned from every student when there were 30 or more students per course. The problems that I have marked with an * are the problems that they are expected to write up and hand in. All problems are presented, but only these are rigorously graded in a written format.

There is also the challenge of trying to make sure that all students get an opportunity to present regularly. I tell students that they are chosen randomly for presentations, but I want them to understand that they can be called on subsequent class days. To make this point, I frequently literally stack the cards for the first couple of class periods. There are students that I know, either from advising or from a previous course, who are more likely to be called to present during the first couple of class periods and there is always a student (or two) who is called twice in the first week.

0.4 Format of the Course

The following is an excerpt of the syllabus, explaining the formatting of the course as I share it with my students.

Attendance Policy: *Students are expected to attend every class. If class must be missed, the student is expected to get the notes from a classmate, and to check the web-page for announcements and updated assignments. The professor will keep a record of attendance.*

Tardiness: *Students are expected to arrive to class on time. If a student is perpetually late, they will be asked to not attend class unless they arrive on time. If tardiness becomes a problem for the class as a whole, people who arrive late will not be permitted to enter the class. If this stricter policy becomes necessary, there will be an announced in class and posted on the web-page.*

Class participation: *Students will be asked to share their solutions to*

problems. During classes, I will randomly call on students to present problems assigned. You may choose to present any problem not yet presented in class. If you are not the first person chosen on a given day, you will not have your choice of all assigned problems. This means that you may not get to present your first choice problem, so you should be prepared with solutions to more than one problem. It is possible that you will be called on to present two days in a row, so you should be constantly working on problems.

You are allowed two “passes” during the course of the semester. These are for days when you have no problems to present. There is no penalty for using these two passes. However, once you have passed twice during the semester, you will not be allowed to pass again, and will be forced to present a problem, so use these wisely.

During student presentations, the rest of the class is encouraged to ask questions, and to think critically about the solution presented by the classmate. The class is responsible for determining the validity of arguments presented, and the instructor will occasionally allow incorrect solutions to stand in class. These incorrectly presented problems will appear on quizzes and tests.

0.5 Assessment of Students

Some people are comfortable gaining a feeling of where their students stand at the end of the semester and using this knowledge of their students to assign grades. I am not comfortable with that, and have therefore talked to a lot of people and stolen a lot of ideas about how to assess my students. I struggle with how to “score” presentations and how much written work to collect.

I also want to make sure that my students are actually learning the definitions, so we have daily quizzes, mostly short, just asking the students to regurgitate two or three definitions. After all, how can they be expected to prove anything without knowing and understanding the definitions?

I have a history, also, of needing to convince my colleagues that students leaving my modified Moore method course were just as prepared for future courses as students leaving a traditional lecture-based proofs course. (I actually think that they are more prepared, but that is another topic.) Because of this, I have kept the traditional exam requirement for the course, and the exams themselves are traditional in both format and content.

All of this leads to a great deal of grading, but I have developed some ways that help me keep this manageable. We will discuss each of these grading issues in turn below.

0.5.1 Homework

In this course, where the students need to learn to write, I feel that I have to grade *some* written work, and have, over the years, fought with myself about how much. “*How much?*” is enough to give them feedback without always feeling like I am buried under papers just from this class. Here is what I’ve come up with. This is what works for me and you should, of course, consider grading less or more depending upon your needs and the needs of your students:

Homework: *Homework will be assigned in class each day. Assignments can be found online at the course website. **Students are not allowed to discuss problems with other members of the class, tutors, or anyone except for the instructor. Students are not to use outside references. This means that students are not allowed to access internet resources or use other textbooks for help. If you have questions about problems, you should see your instructor.***

Many problems will be assigned and all problems will be discussed in class. However, only problems marked with an asterisk () will be collected and graded. For those to be collected, see the standards below.*

Every problem completed should be written on a separate sheet of paper. Problems can be submitted prior to presentations or after presentations. Problems will be collected only at the beginning of every class, and will be graded in the following way:

10 pts - Correct work, turned in pre-presentation

1 - 9 pts - Almost correct work, turned in pre-presentation

5 pts - Correct work, turned in post-presentation

Incorrect solutions to problems which are submitted after a problem has been presented in class will not receive any points. Problems can be resubmitted if you are unhappy with your score up to three class periods after the problem has been presented in class.

Notice that this last works most easily for a Monday-Wednesday-Friday course - in this case, three class periods is one week. It is a little harder for you and for the students to keep track of if you have a Tuesday-Thursday course meeting, but I have done it. In this case, it is especially helpful to have a course website to help your students (and yourself) keep track of the deadlines.

0.5.2 Quizzes

As mentioned above, I feel the need to make sure that my students are actually learning the vocabulary required for the course. To do this, I give very short daily quizzes. The syllabus says:

There will be a short quiz given in class every day. Most days these will be vocabulary quizzes, but might also ask for examples, counterexamples, or short proofs.

I do **not** think that these quizzes are necessary for the students' success in the course. However, I find that my students are more likely to memorize the definitions (as opposed to looking them up every time they work a problem) when I give these daily quizzes. Actually, these take little time to grade, and so I think they are worth the extra effort.

0.5.3 Presentations

This is the hardest part for me to evaluate objectively. I wanted to have a scoring system in place that I could explain to students so that they would understand the basis of their grade, and so that I felt like I understood objectively where they should be. The following is the relevant excerpt from my syllabus:

Class participation: *Students will be asked to share their solutions to problems. During classes, I will randomly call on students to present problems assigned. You may choose to present any problem not yet presented in class. If you are not the first person chosen on a given day, you will not have your choice of all assigned problems. This means that you may not get to present your first choice problem, so you should be prepared with solutions to more than one problem. It is possible that you will be called on to present two days in a row, so you should be constantly working on problems.*

You are allowed two "passes" during the course of the semester. These are for days when you have no problems to present. There is no penalty for using these two passes. However, once you have passed twice during the semester, you will not be allowed to pass again, and will be forced to present a problem, so use these wisely.

During student presentations, the rest of the class is encouraged to ask questions, and to think critically about the solution presented by the classmate. The class is responsible for determining the validity of arguments presented, and the instructor will occasionally allow incorrect solutions to stand in class. These incorrectly presented problems will appear on quizzes and tests.

Presentations will be scored in the following way:

- Accuracy of the problem you present, including following guidelines

below.

- *Defense of your work, including, including following guidelines below.*

Points: You will be awarded a “point” in the appropriate category every time you contribute in one of the following ways:

- *P - presentation points awarded*
 - 5 pts** - Correct presentation
 - 2-4 pts** - Presentation with error(s)
- *Q - asking a **good** question of the presenter*
- *C - an oral contribution other than the two categories above*
- *I - contributing a demonstration of mathematical insight*

Remember that the presenter will always have the first chance to answer a question.

Guidelines For Your Presentation:

- *Write the problem on the board.*
- *State what method/theorem/idea you will use.*
- *Clearly explain each step.*
- *Do not use “stupid”, “trivial”, “obvious”, etc.*

Guidelines For Defending Your Work:

- *You must answer you classmates’ and professor’s questions in a respectful manner.*
- *Do not use “stupid”, “trivial”, “obvious”, etc.*
- *You must try to answer every question posed.*
- *It’s OK to say, “I’m not sure that I understand your question.” It is not OK to say, “Your question doesn’t make sense.”*
- *Talk to the class, not to the board.*

Guidelines For Criticism of Classmates’ Work:

- *You are to ask questions about your classmates’ work. Do NOT suggest another technique. In some cases, there may be more than one way to solve a problem.*
- *Do not use “stupid”, “trivial”, “obvious”, etc.*
- *You must ask questions in a respectful manner.*
- *It’s OK to say, “Can you explain how you got from line 3 to line 4?” It is not OK to say, “Line 4 is wrong,” or “Line 4 doesn’t make sense.”*

0.5.4 Exams

As mentioned above, I give fairly traditional exams during this course. I give two mid-semester exams and a final exam. Here are the relevant portions of the syllabus for exams:

There will be two exams during the semester. They are tentatively scheduled to occur on: (insert days here). Any changes to this schedule will be announced in class and posted on the web-page.

If a student misses an exam, the student will be allowed to replace that exam score with their score on the final exam if the student contacts the instructor prior to the exam and the student takes and passes the next exam at the regularly scheduled time.

*The final exam will be held over two days, with certain topics (to be announced later) on each day. The final exam will be held on **the last two class periods** during out normal classtime. It will be comprehensive.*

This last is a new development for me, but I thought it worked very nicely. The exam was broken into two pieces - the first of which was definitions, true/false problems, and a couple of proofs that I told them ahead of time they should understand well (or memorize). These proofs were of “basic” facts like the existence of an infinite number of primes, the fact that $\sqrt{2}$ is irrational, and that every natural number greater than one is either prime or the product of primes.

The second day of the final exam is the day of proofs. The students are asked to prove or disprove a variety of statements, some of which are old homework problems, some of which use the old definitions to prove new things, and one or two of which provide them with a new definition and ask them to prove something new from that new definition. Each part of the final is worth 50 points, and the sum gives their exam score.

0.5.5 Grading Plan

When I began using the Moore Method, I got much advice on how much weight to assign to the presentation portion of the course. I have done this in many different ways over the past several years, but there is some universal advice that I think bears repeating - even if you will have a homework component and exams, you need to weight the presentation portion of the course enough so that students will take it seriously. For me, that means having the presentation portion worth between 20% and 30% of their course grade. In semesters where the presentations are worth only 10% or 15%, the students are more likely to come to class unprepared, and their course grade is not really indicative of how prepared they were on a daily basis for presentations and discussion.

Most recently, the course grade was assigned by weighting each of the

following categories evenly, for 20% of the final grade each: Homework and Quizzes, Class Participation, Exam 1, Exam 2, and the Final Exam.

0.5.6 Assigning Grades

I find one of the hardest parts of my job is assigning grades. Since many students will hand in many problems after they are presented, it is not uncommon for their averages to be in the 50s or 60s for their homeworks. I do not score this on a standard scale, but usually curve, taking their score to be a percentage of the maximum number of points scored by any student.

Similarly, it is hard to know what is a “good” presentation score. The first thing I do here is to find their percentage of points compared to the person who has the highest number of points. From here, I look for natural breaks in the numbers to assign more traditional “A”, “B”, “C” grades to their presentation scores, which I then use in calculating their class grades based on the percentages above.

0.6 Other Helpful Information

I *never* tell my students at the beginning that this course is being run via Moore Method, or indicate in any way that what they are doing in my class is different than happens in any other proof course. I do tell them that their job for the semester is to learn to write and to critically evaluate their own proofs and the proofs presented by their classmates. I reiterate this many many times over the course of the semester. As the semester continues, I mention that this is something called “Inquiry-based learning” or “student-centered learning” or “The Moore Method.” By then, however, they understand both their role and mine and understand the methodology of the course without me having to sell them on it.

0.7 Acknowledgements and Thanks

Many of the problems in this collection are common to many textbooks, however the books that I referenced most often are the following:

- Burger, Edward. *Extending the Frontiers of Mathematics: Inquiries into proof and argumentation*, Key College Publishing, 2007.
- Devlin, Keith. *Sets, Functions, and Logic: An Introduction to Abstract Mathematics, Third Edition*, Chapman & Hall/CRC, 2004.
- Fletcher, Peter and Patty, C. Wayne. *Foundations of Higher Mathematics, Second Edition*, PWS-Kent Publishing Company, 1992.

- Schumacher, Carol. *Chapter Zero: Fundamental Notions of Abstract Mathematics, Second Edition*, Addison Wesley, 2001.
- Smith, Douglas; Maurice Eggen, and Richard St. Andre. *A Transition to Advanced Mathematics, Sixth Edition*, Thomson/Brooks Cole, 2006.

Other problems were gotten from the Spanish-language movie

L'habitación de Fermat (Fermat's Room) (2007). They are cited as appropriate in the text.

I want to express thanks to my mentors when I started using the modified Moore method - there are too many to name, but W. Ted Mahavier and Brian Raines deserve special mention for the many many emails that they were willing to exchange with me. More recently, thanks must be given to Carol Schumacher, Ron Taylor, and Stan Yoshinobu, for their many helpful discussions of teaching issues and philosophy - my teaching continues to evolve because of conversations with you, and I am so grateful for your time and input.

To the Student

Your role in this course is to solve the following problems. Feel free to work out of order - sometimes the problems you will understand and be able to complete first will happen later in the problem sequence. However, when working on later problems, remember things that have already been proven - sometimes they provide hints about how to proceed with later problems.

The purpose of these notes is to guide you on a path towards critical thinking - by the end of the course, you should be able to write clear and concise solutions to the problems posed (it is your instructor's job to provide you with feedback about your writing), be able to critically examine solutions posed by other students, and, therefore, critically evaluate your own solutions. This last is the most difficult of all - it is easy to convince yourself that your solution works, but will it convince others? Having a critical eye for your own work is a necessary and challenging skill to develop.

One of the best skills that you can learn as a mathematician is to ask good questions. For this course, you should be asking good questions of your instructor, your colleagues (in the appropriate forum as indicated by your instructor), and (mostly importantly) yourself.

Good luck and enjoy!

Chapter 1

Introduction - Thinking Outside the Box

Solve the following problems by any means necessary. Solutions to these nine problems are due at the beginning of class on *(Insert date of second class here)*. Solutions should be written clearly, professionally, and in complete sentences in such a way that they are complete and explain the answer to the question.

1. There are three boxes of candy. One box contains mint candies, one chocolate candies, and the other is mixed. All three boxes are incorrectly labeled. What is the smallest number of candies that you need to remove and sample to be able to correctly label all three boxes? ¹
2. A sealed room contains one light bulb. Outside of the room, there are three switches, only one of which operates the bulb. You are outside the room, able to operate the switches in any way you see fit, but when the door is opened for the first time, you must determine which switch operates the light. What do you do? ²
3. Describe how one can use a four-minute hourglass and a seven-minute hourglass to measure a period of nine minutes. ³
4. Two doors are guarded by two men, one of whom always lies and one of whom always tells the truth; however, you do not know which man is which. One of the doors leads to freedom and one to captivity. Determine a single question that, if asked of one of the guards, would reveal the door to freedom with certainty.
5. A cruel calculus instructor decided to terrorize her students. The instructor announced that during the next class, the students will line up, facing away from the front of the line. The instructor will then place either a white or a gray dunce cap on each student's head. Each student

¹ibid.

²ibid.

³ibid.

will be unable to see his or her own cap but will be able to see the cap colors of all those classmates who are in front of him or her.

Starting at the head of the line, each student will be asked, in turn, “What is the color of your dunce cap?” Students will only be allowed to respond by saying “white” or “gray.” The students who answer correctly will be given A’s and the student who answer incorrectly will fail the course. For each student, the instructor will respond with either “Correct! You receive an A” or “you fail! Get out of my classroom, you dunce,” depending on the correctness of the answer. All students will be able to hear all students’ answers and the instructor’s responses. Knowing this horrific fate that awaits them, the students have all night to come up with a plan. If there are n students in the class, how many of them can be guaranteed to receive A’s? Your challenge is to devise a scheme that the students can employ to allow as many of them as possible to receive A’s. (Caution: No “cheating” is allowed; that is, students cannot use the tones of their voices or say additional phrases or use hand gestures to provide any additional information.)⁴

6. Consider the following mathematical illusion: A regular deck of 52 playing cards is shuffled several times by an audience member until everyone agrees that the cards are completely shuffled. Then, without looking at the cards themselves, the magician divides the deck into two equal piles of 26 cards. The magician taps both piles of face-down cards three times. Then, one by one, the magician reveals the cards of both piles. Magically, the magician is able to have the cards arrange themselves so that the number of cards showing black suits in the first pile is identical to the number of cards showing red suits in the second pile. Your challenge is to figure out the secret to this illusion and then perform it for your friends.⁵
7. Some number of coins are spread out on a table. They lie either heads up or tails up. Unfortunately you are blindfolded and thus both the coins and the table upon which they sit are hidden from view. Certainly you can feel your way across the table and count the total number of coins on the table’s surface, but you cannot determine if any individual coin rests heads up or down (perhaps you are wearing gloves). You are informed of one fact (beyond the total number of coins on the table): Someone tells you the number of coins that are lying heads up. You can now rearrange the coins, turn any of them over, and move them in any way you wish, as long as the final configuration has all the coins resting (heads or tails up) on the table. Your challenge is to turn over whatever coins you wish and divide the coins into two collections so

⁴Burger

⁵ibid.

that one collection of coins contains the same number of heads up coins as the other collection contains.⁶

8. You find yourself on a reality TV show that has you completing with other real people in totally artificial circumstances. In one scenario, you are given nine balls of clay. You are informed by the program's B-celebrity host that hidden inside one of those clay balls is a key that will unlock a refrigerator that houses a vast quantity of food. Since the producers "thought" the ratings would be higher if the contestants were deprived of nutrition, even the thought of brussels sprouts makes your mouth water. You are told that the eight balls that do not contain the key to your dietary dreams all weigh the same. The special ball with the key insider weighs slightly more, but not enough for you to feel the difference by holding the balls in your hand. One of the program's sponsors, *Replace-Oh!*, the company that manufactures one-time-use balance scales (with the slogan "Weigh aweigh then throw away!"), has agreed to provide some of its scales in exchange for a few shameless plugs throughout the program. Their scales will tell which side is heavier and then instantly self-destruct. You are only allowed to break open one clay ball to see if you can find the refrigerator key. Your challenge is to determine the fewest disposable balance scales required to guarantee that you can identify the ball with the key. Justify your answer.⁷

⁶ibid.

⁷ibid.

Chapter 2

Introduction to Number Theory

Definition 1. A natural number is a counting number, as in $1, 2, 3, \dots$

Definition 2. A whole number is a counting number or zero, as in $0, 1, 2, 3, 4, \dots$

Definition 3. An integer is a positive or negative counting number or zero, as in

$$\dots, -2, -1, 0, 1, 2, 3, \dots$$

9. (*) Prove there does not exist an equilateral triangle in the plane whose vertices are at integer lattice points (x, y) .

Definition 4. An integer n is even if there exists an integer k so that it can be written as $n = 2k$.

Definition 5. An integer n is odd if there exists an integer k so that it can be written as $n = 2k + 1$.

10. (*) Prove that there exist integers m and n so that $2m + 7n = 1$.
11. (*) Prove that there do not exist integers m and n so that $2m + 4n = 7$.
12. (*) Let a and b be two integers. Prove that if $a + b$ is even then $a - b$ is even.
13. (*) Prove that an integer n that can be written as $n = 2k - 1$ for some integer k is odd.
14. Prove that the product of two even numbers is even.
15. Prove that the product of an even number and an odd number is even.
16. Prove that the product of two odd numbers is odd.

Chapter 3

Logic

Definition 6. *A mathematical statement is a declarative sentence which is either true or false.*

Notice that a mathematical statement does not need to be a statement about mathematics.

17. Which of the following are mathematical statements? Explain
- (a) $2^3 + 3^2 = 17$
 - (b) $8x^3 + 6x^2 - 4x + 2$
 - (c) Broccoli is green.
 - (d) Will you marry me?
 - (e) It is a beautiful day.
18. Which of the following are **true** mathematical statements? Which are false? Explain.
- (a) Broccoli is a vegetable and $3 < \sqrt{17}$.
 - (b) Mercury is the closest planet to the sun and $3 > \sqrt{17}$.
 - (c) Shakespeare was a playwright or $|3| = |-3|$.
 - (d) Chile is a country in South America or 8 is a prime number.
 - (e) 7 is an odd number or $\pi > 3$.
19. The problem above asks about two types of statements: “and” statements and “or” statements. What is necessary for an “and” statement to be true? For an “or” statement to be true?
20. (*) Suppose each of the following three statements is true:
John is smart.
John or Mary is ten years old.
If Mary is ten years old, then John is not smart.
Which of the following statements are true and why?

- (a) Mary is 10 years old.
- (b) John is 10 years old.
- (c) Either John or Mary is not 10 years old.

Definition 7. The negation of a statement is a statement having the opposite truth value of the original statement. In other words, if the original statement is true, the negation is false; and if the original statement is false, the negation is true.

Example 8. Original Statement: $3 > \pi$.

Silly negation: 3 is not greater than π .

Useful Negation: $3 \leq \pi$.

Example 9. Original Statement: Curtis is nice.

Silly negation: It's not true that Curtis is nice.

Useful Negation: Curtis is not nice.

Notice that the negation is not "Curtis is mean." That statement is stronger than the statement that he is not nice.

21. (*) Find grammatically correct negations of each of the following statements:
 - (a) The number x is negative.
 - (b) For all x , $f(x) > 0$.
 - (c) There exists an x such that $g'(x)$ is undefined.
22. Write a useful negation of each of the following statements:
 - (a) All cows eat grass.
 - (b) There is a horse that does not eat grass.
 - (c) There is a car that is blue and weighs less than 4000 pounds.
 - (d) Every math book is either white or hard to read.
 - (e) Some cows are spotted.
 - (f) No car has 15 cylinders.

Definition 10. A truth table is a way of keeping track of the truth or falsity of a complicated statement for all possible truth values of the component statements.

Example 11. ¹ Let P and Q be statements. Then the statement " P and Q " has the following truth values, depending upon the truth values of P and of Q :

P	Q	P and Q
T	T	T
T	F	F
F	T	F
F	F	F

Definition 12. Two statements are equivalent if their truth table match for all possible values of the component statements.

23. (*) Express the statement “Not (A and B)” as an equivalent statement involving only “not” and “or.” Then express the statement “Not (A or B)” as an equivalent statement involving only “not” and “and.” Justify your answers with truth tables.

Definition 13. A conditional statement is one of the form “If P then Q.”

24. Which of the following conditional statements are true and why?

- (a) If $1 + 1 = 3$ then $\sqrt{36} = -6$.
- (b) If $\cos \pi = 0$ then $\sin \pi = 0$.
- (c) If $\sin \pi = 0$ then $\cos \pi = 0$.
- (d) If $x = -3$ satisfies $x^2 = 9$ then $\log_2 \frac{1}{8} = -3$.

25. (*) Complete the rest of the truth table for a conditional statement “If P, then Q.”

P	Q	If P, then Q
T	T	
T	F	
F	T	
F	F	

26. Find the negation of the statement “If $\sin \pi = 0$, then $\cos \pi = 0$.”
27. (*) What is the negation of the statement “If P, then Q”? Justify your answer with a truth table.

Definition 14. The converse of the conditional statement “If P, then Q” is “If Q, then P.”

Definition 15. The inverse of the conditional statement “If P, then Q” is “If not P, then not Q.”

Definition 16. The contrapositive of the conditional statement “If P, then Q” is “If not Q, then not P.”

28. (*) Consider an implication, its converse, its inverse, and its contrapositive. Which of these statements are equivalent to one another? Justify your answers with truth tables.
29. The converse of the inverse is another conditional statement. Which one? Explain.

30. Give an example of a true conditional statement with a false converse, or explain why no such example can exist. Explain.
31. Give an example of a true conditional statement with a false contrapositive, or explain why no such example can exist. Explain.
32. (*) Give an example of a false conditional statement with a false inverse, or explain why no such example can exist. Explain.

Chapter 4

More Number Theory and Basic Proof Techniques

Definition 17. A number n is a rational number if it can be written as $\frac{p}{q}$ for some integers p and q where $q \neq 0$.

Fact. If we assume that $q > 0$, every rational number can be written uniquely in a reduced form (i.e., where p and q have no common divisors).

33. (*) Prove that $\sqrt{2}$ is not a rational number. ²

Definition 18. A natural number n is prime if it has exactly two divisors, namely 1 and itself.

34. Prove or disprove: If n is prime, then $2^n - 1$ is prime.
35. (*) Prove or disprove: If n is a natural number, $n^2 + n + 41$ is prime.
36. (*) Prove or disprove: The square of an odd integer is odd.

Definition 19. If a and b are integers, a divides b if there exists an integer k so that $b = ak$.

37. Prove or disprove: The product of two even integers is divisible by 4.
38. Prove or disprove: Let a and b be integers. If a divides b then a divides bc for all integers c .
39. Prove or disprove: Let a and b be integers. If a divides b and a divides c , then a divides $b - c$.
40. (*) Prove or disprove: Let a and b be integers. If a and b are both positive and a divides b and b divides a then $a = b$. Does this still hold if a and b are not necessarily positive? Why or why not?

41. Prove or disprove: Let a, b , and c be integers. If ab divides c then a divides c .
 42. Prove or disprove: Let a, b , and c be integers. If ac divides bc then a divides b .
 43. (*) Prove or disprove: An integer n is even if and only if n^2 is even.
 44. Prove or disprove: If n is a natural number, then $n^2 + n$ is even.
- Fact.** Every integer other than 1 and -1 has a prime divisor.
45. (*) Prove or disprove: There are an infinite number of primes.
 46. (*) Let a, b and c be integers such that $a^2 + b^2 = c^2$. Prove at least one of a and b is even.

Definition 20. For a natural number n , $n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1$.

47. Prove that if n is a natural number greater than 1, then $n! + 1$ is odd.
48. How many pairs of primes p and q are there such that $q - p = 3$? Prove your answer.
49. Prove or disprove: Let n be an integer. If 3 divides n^2 , then 3 divides n .
50. Prove or disprove: $\sqrt{3}$ is irrational.

Definition 21. Let a and b be integers. The greatest common divisor of a and b is denoted by $\gcd(a, b)$ and is the natural number d that satisfies the following two conditions:

- (i) d divides both a and b
- (ii) If n is an integer that divides both a and b then n divides d .

Example 22. Notice that for any integers a and b , 1 is a common divisor. Therefore, the greatest common divisor always exists.

51. Let b be a nonzero integer. Prove that $\gcd(0, b) = |b|$.

Axiom: The Division Algorithm - If a and b are positive integers with $b \leq a$, then there exists a natural number q (called the quotient) and a nonnegative integer r (called the remainder) such that $a = bq + r$ where $0 \leq r < b$.

Example 23. If $a = 7$ and $b = 3$ then $q = 2$ and $r = 1$ so that $7 = 3 \cdot 2 + 1$.

If $a = 20$ and $b = 5$ then $q = 4$ and $r = 0$ so that $20 = 5 \cdot 4 + 0$.

Be careful - the remainder needs to be smaller than b so while it is true that $7 = 3 \cdot 1 + 4$, this is not the way we want to choose q and r .

Axiom: The Euclidean Algorithm - Let a and b be two positive integers with $b \leq a$. Let d be the $\gcd(a, b)$. Then the following two statements hold:

(i) There are two lists of positive integers q_i and r_i such that

$$b > r_1 > r_2 > r_3 > \cdots > r_{k-1} > r_k > r_{k+1} = 0 \text{ and}$$

$$a = bq_1 + r_1$$

$$b = r_1q_2 + r_2$$

$$r_1 = r_2q_3 + r_3$$

$$\vdots$$

$$r_{k-3} = r_{k-2}q_{k-1} + r_{k-1}$$

$$r_{k-2} = r_{k-1}q_k + r_k$$

$$r_{k-1} = r_kq_{k+1} \text{ (that is, } r_{k+1}=0\text{)}$$

and $d = r_k$.

(ii) The GCD of a and b may be written as an integer combination of a and b ; that is, there exist integers x and y so that $d = ax + by$.

Example 24. We can use the Euclidean Algorithm to find $\gcd(44, 104)$.

$$104 = 44(2) + 16$$

$$44 = 16(2) + 12$$

$$16 = 12(1) + 4$$

$$12 = 4(3)$$

The last non-zero remainder is the greatest common divisor, and so $\gcd(44, 104) = 4$.

52. Use the Euclidean Algorithm to find $\gcd(219, 69)$.

53. Find integers m and n so that $\gcd(219, 69) = 219m + 69n$.

54. Use the Euclidean Algorithm to find $\gcd(10245, 5357)$.

55. Find integers m and n so that $\gcd(10245, 5357) = 10245m + 5357n$.

Definition 25. Two integers a and b are relatively prime if

$$\gcd(a, b) = 1.$$

Comment. Notice that **prime** is a quality of one number. **Relatively prime** is a quality of a pair of numbers.

56. What does the Euclidean Algorithm tell us about integers a and b with $\gcd(a, b) = 1$? Explain. (Notice that saying that they are relatively prime does NOT count.)
57. (*) Prove or disprove: If a, b and c are integers such that a and b are relatively prime and $a \mid bc$ then $a \mid c$.
58. Prove or disprove: If p is a prime and a is an integer such that p does not divide a then a and p are relatively prime.
59. Prove or disprove: If p is a prime and a is an integer, then $\gcd(p, a) = p$ if and only if p divides a .
60. (*) Prove or disprove: Let a and b be integers and let p be a prime number. If $p \mid ab$ then $p \mid a$ or $p \mid b$.
61. (*) Prove or disprove: Let a, b and c be integers such that $\gcd(a, c) = \gcd(b, c) = 1$. Prove that $\gcd(ab, c) = 1$.
62. (*) Prove or disprove: Let a, b and c be integers such that a and b are relatively prime and c divides $a + b$. Prove that $\gcd(a, c) = \gcd(b, c) = 1$.

Definition 26. Let a and b be positive integers. A natural number n is the least common multiple of a and b if it satisfies the following two properties:

- (i) a divides n and b divides n
- (ii) If m is any other number so that a divides m and b divides m then n divides m .

Comment: Notice that if ab is always a common multiple of a and b and, therefore, the $\text{lcm}(a, b)$ always exists.

63. Prove or disprove: Let a and b be natural numbers and $\text{lcm}(a, b) = m$. Then $m = b$ if and only if a divides b .
64. (*) Prove or disprove: Let a and b be natural numbers and $\text{lcm}(a, b) = m$. Then $\text{lcm}(an, bn) = mn$ for all natural numbers n .
65. (*) Prove or disprove: Let a and b be natural numbers with $\gcd(a, b) = d$, and $\text{lcm}(a, b) = m$. If $d = 1$ then $m = ab$, i.e. if a and b are relatively prime then their least common multiple is the product of a and b .
66. (*) Let d and n be nonzero integers. Prove or disprove: If $d \mid n^2$ then $d \mid n$.
67. Prove or disprove: Let n be an integer and p be prime. Prove that if $p \mid n^2$ then $p \mid n$.

68. Prove that if p is a prime, then $\sqrt{p}$ is irrational.
69. Prove or disprove: The product of any three consecutive integers is a multiple of three.
70. Extend the previous problem and prove your extension.

Chapter 5

Induction

Axiom: If we want to show that a statement is true for all natural numbers, we can show that each natural number makes the statement true. However, since the set of natural numbers is infinite, we need to be clever about this. It is enough to prove a statement using the Principle of Mathematical Induction. This involves three steps:

- (a) Show that the statement is true for the smallest natural number, 1 (or some other base case, if relevant).
- (b) Assume that the statement holds for a certain natural number, k .
- (c) Show that whenever the statement is true for a certain natural number, k , it is then true for the next natural number, $k + 1$. This shows that it is true for every natural number starting at a specific point (usually, but not always, 1).

Example 27. We wish to show that $1 + 3 + 5 + \dots + (2n - 1) = n^2$ for every natural number n .

Proof: We will first show that this holds for a base case (where $k = 1$).

If $n = 1$ this statement claims that $1 = 1^2$. The right hand side is equal to one, so this is a true statement.

We next assume that this is true for some natural number k and we show that this is true for the next natural number $k + 1$.

Therefore, we assume that $1 + 3 + 5 + 7 + \dots + (2k - 1) = k^2$ and we want to show that

$$1 + 3 + 5 + 7 + \dots + (2k - 1) + (2(k + 1) - 1) = (k + 1)^2.$$

In other words, we want to see that

$$1 + 3 + 5 + 7 + \dots + (2k - 1) + (2k + 2 - 1) = (k + 1)^2,$$

or

$$1 + 3 + 5 + 7 + \dots + (2k - 1) + (2k + 1) = (k + 1)^2.$$

Let us begin by examining that left hand side of the desired equation:

$$1 + 3 + 5 + 7 + \cdots + (2k - 1) + (2k + 1) =$$

Noticing that the first part of this sum is part of our assumption statement, we group that part together:

$$(1 + 3 + 5 + 7 + \cdots + (2k - 1)) + (2k + 1) =$$

and by substituting from our assumption statement, $1 + 3 + 5 + 7 + \cdots + (2k - 1) = k^2$, we get:

$$k^2 + 2k + 1 = (k + 1)^2$$

as desired. Therefore, we have shown that the desired statement is true for $k = 1$, and that whenever it is true for one natural number, it is true for the next natural number. We have, therefore, shown that it is true for all natural numbers.

71. Prove or disprove: The sum of the first n natural numbers is $\frac{n(n+1)}{2}$, i.e.

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

72. (*) Prove or disprove: For every natural number n ,

$$2 + 4 + 6 + \cdots + 2n = n(n+1).$$

73. Prove or disprove: For every natural number n ,

$$3 + 6 + 9 + \cdots + 3n = \frac{3n(n+1)}{2}.$$

74. Prove or disprove: For every natural number n ,

$$4 + 8 + 12 + \cdots + 4n = 2n(n+1).$$

75. (*) Prove or disprove: For every natural number n ,

$$5 + 10 + 15 + \cdots + 5n = \frac{5n(n+1)}{2}.$$

76. Prove or disprove: For all natural numbers a it is true for every natural number n that,

$$a + 2a + 3a + \cdots + na = \frac{na(n+1)}{2}.$$

77. Prove or disprove: For every natural number n ,

$$2^0 + 2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 1.$$

78. Prove or disprove: For every natural number n ,

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

79. (*) Prove or disprove:: For every natural number n ,

$$1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2.$$

80. Prove or disprove:: For every natural number n ,

$$\left(1 + \frac{1}{1}\right) \left(1 + \frac{1}{2}\right) \cdots \left(1 + \frac{1}{n}\right) = n + 1.$$

81. Prove or disprove: For every natural number n ,

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{n^2(n+1)^2}{4}.$$

82. (*) Prove or disprove:: For every natural number n ,

$$1^3 + 3^3 + 5^3 + \cdots + (2n-1)^3 = n^2(2n^2 - 1).$$

83. (*) Prove or disprove: $3 \mid (n^3 + 2n)$ for every natural number n .

84. Prove or disprove: $4 \mid (13^n - 1)$ for every natural number n .

85. (*) Prove or disprove: For every natural number n , 6 divides $n^3 - n$.

86. Prove or disprove: The product of any three consecutive natural numbers is divisible by 6.

87. (*) Prove or disprove: Every natural number greater than 1 is either prime or the product of prime numbers.

88. Prove or disprove: For every natural number n ,

$$3 + 6 + 12 + \cdots + 3(2^{n-1}) = 3(2^n - 1).$$

89. (*) Prove or disprove: For every natural number n , $9^n - 8n - 1$ is divisible by 64.

Chapter 6

Set Theory

Definition 28. A set is a specified collection of objects.

Example 29. The set of integers between -1 and 4 , including the endpoints, is $\{-1, 0, 1, 2, 3, 4\}$.

Definition 30. If A is a set and x is an object that belongs to A then x is an element of A , denoted $x \in A$. If x is not an object that belongs to A , we say that x is not an element of A (or is not a member of A), denoted $x \notin A$.

90. List all of the elements in the following sets:

- (a) The set of natural numbers strictly less than 6.
- (b) The set of integers whose square is less than 17.
- (c) The set of prime numbers less than 100.
- (d) The set of rational numbers strictly between 0 and 1.

Definition 31. If A and B are sets, B is a subset of A , denoted $B \subseteq A$ if every member of B is a member of A .³

91. True or False, and explain your reasoning. $\mathbb{N}$ is the set of natural numbers, $\mathbb{Q}$ is the set of rational numbers, $\mathbb{Z}$ is the set of integers, and $\mathbb{R}$ is the set of real numbers.

- (a) $\mathbb{N} \subseteq \mathbb{Q}$
- (b) $\mathbb{Z} \subseteq \mathbb{N}$
- (c) $\mathbb{Q} \subseteq \mathbb{Z}$
- (d) $\mathbb{N} \subseteq \mathbb{R}$
- (e) $\mathbb{R} \subseteq \mathbb{Q}$
- (f) $(6, 9] \subseteq [6, 10)$
- (g) $[7, 10] \subseteq \mathbb{R}$

Definition 32. Sets A and B are equal, denoted $A = B$, if $A \subseteq B$ and $B \subseteq A$.

Definition 33. The set with no members is called the empty set and denoted $\emptyset$.

92. True or False and explain your reasoning.

- (a) $\emptyset \subseteq \mathbb{N}$
- (b) $\emptyset \in \mathbb{N}$
- (c) $\emptyset \in \{\emptyset, \{\emptyset\}\}$
- (d) $\emptyset \subseteq \{\emptyset, \{\emptyset\}\}$
- (e) $\{\emptyset\} \subseteq \{\emptyset, \{\emptyset\}\}$
- (f) $\{\emptyset, \{\emptyset\}\} \subseteq \{\{\emptyset, \{\emptyset\}\}\}$

93. True or False and explain your reasoning.

- (a) For every set A , $\emptyset \subseteq A$
- (b) For every set A , $\emptyset \in A$

94. Give an example, if there is one, of sets A , B , and C such that the following are true. If there is no example, state such. Explain your reasoning on all problems.

- (a) $A \subseteq B, B \not\subseteq C$ and $A \subseteq C$.
- (b) $A \subseteq B, B \subseteq C$, and $C \subseteq A$.
- (c) $A \not\subseteq B, B \not\subseteq C$ and $A \subseteq C$.
- (d) $A \subseteq B, B \not\subseteq C$ and $A \not\subseteq C$.

95. (*) Let A be a set. Prove or disprove: Then $\emptyset \subseteq A$ and $A \subseteq A$.

96. (*) Let A , B and C be sets. Prove or disprove: If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$.

Definition 34. A subset $A \subseteq B$ is a proper subset if $A \subseteq B$ and $A \neq B$. This is denoted by $A \subset B$.

97. List all of the subsets of the following sets. Which ones are proper?

- (a) $\emptyset$
- (b) $\{1\}$
- (c) $\{1, 2\}$
- (d) $\{\{\emptyset\}\}$
- (e) $\{\emptyset, \{\emptyset\}\}$

Definition 35. The power set of a set A is the set of all (proper and not proper) subsets of A . This is denoted $\mathcal{P}(A)$.

98. True or False and explain your reasoning.

- (a) $\emptyset \in \mathcal{P}(\{\emptyset, \{\emptyset\}\})$
- (b) $\{\emptyset\} \in \mathcal{P}(\{\emptyset, \{\emptyset\}\})$
- (c) $\{\{\emptyset\}\} \in \mathcal{P}(\{\emptyset, \{\emptyset\}\})$
- (d) $\emptyset \subseteq \mathcal{P}(\{\emptyset, \{\emptyset\}\})$
- (e) $\{\emptyset\} \subseteq \mathcal{P}(\{\emptyset, \{\emptyset\}\})$
- (f) $\{\{\emptyset\}\} \subseteq \mathcal{P}(\{\emptyset, \{\emptyset\}\})$

99. True or False and explain your reasoning.

- (a) $3 \in \mathbb{Q}$
- (b) $\{3\} \subseteq \mathcal{P}(\mathbb{Q})$
- (c) $\{3\} \in \mathcal{P}(\mathbb{Q})$
- (d) $\{\{3\}\} \subseteq \mathcal{P}(\mathbb{Q})$
- (e) $\{3\} \subseteq \mathbb{Q}$
- (f) $\{\{3\}\} \in \mathcal{P}(\mathbb{Q})$

100. (*) Prove or disprove: Let n be a natural number and A be a set containing n elements. The number of elements in $\mathcal{P}(A)$ is 2^n .

101. Give an example, if there is one, of each of the following. If there is no example, state such. Explain your reasoning on all problems.

- (a) A set A such that $\mathcal{P}(A)$ has 64 elements.
- (b) Sets A and B such that $A \subseteq B$ and $\mathcal{P}(B) \subseteq \mathcal{P}(A)$.
- (c) A set A such that $\mathcal{P}(A) = \emptyset$
- (d) A set A such that $\mathcal{P}(A) = \{\emptyset\}$
- (e) Sets A , B , and C such that $A \subseteq B$, $B \subseteq C$ and $\mathcal{P}(A) \subseteq \mathcal{P}(C)$.

102. Find three sets A , B , and C such that $A \in B$, $B \in C$, and $A \in C$.

For all of the following, A , B , and C are sets.

103. True or False and explain your reasoning.

- (a) The empty set is a proper subset of every set.
- (b) If A is a proper subset of $\emptyset$, then $A = \{17\}$.
- (c) If $A \subseteq B$ then $A = B$.
- (d) If $A = B$ then $A \subseteq B$.
- (e) Since $\emptyset$ is a member of $\{\emptyset\}$, $\emptyset = \{\emptyset\}$.
- (f) There is a set that is a member of every set.
- (g) There is a set which is a member of every power set.

Definition 36. If A and B are sets, then the union of A and B is the set of all objects that belong to A or belong to B , denoted $A \cup B$. In other words,

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

Definition 37. If A and B are sets, then the intersection of A and B is the set of all objects that belong to both A and B , denoted $A \cap B$. In other words,

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

104. Prove or disprove: $\emptyset \cap A = \emptyset$ and $\emptyset \cup A = A$

105. (*) Prove or disprove: $A \cap B \subseteq A$

106. (*) Prove or disprove: $A \subseteq A \cup B$

107. Prove or disprove: $A \cup B = B \cup A$ and $A \cap B = B \cap A$

108. Prove or disprove:

$$A \cup (B \cap C) = (A \cup B) \cap C$$

and

$$A \cap (B \cup C) = (A \cap B) \cup C.$$

109. Prove or disprove: $A \cup A = A = A \cap A$

110. (*) Prove or disprove: If $A \subseteq B$, then $A \cup C \subseteq B \cup C$ and $A \cap C \subseteq B \cap C$.

Definition 38. Let A and B be sets. Then the complement of A relative to B is the set $\{x \in B \mid x \notin A\}$, denoted $B - A$.

Definition 39. If for a certain problem, all of the sets being considered are subsets of a given set U then U is called a universal set.

Definition 40. If U is the universal set and $A \subseteq U$ then the complement of A relative to U is denoted A' and is $U - A = A' = \{x \in U \mid x \notin A\}$

111. Prove or disprove: $(A')' = A$
 112. Prove or disprove: $(A \cup B)' = A' \cap B'$
 113. Prove or disprove: $(A \cap B)' = A' \cup B'$
 114. (*) Prove or disprove: $A - B = A \cap B'$
 115. (*) Prove or disprove: $A \subseteq B$ if and only if $B' \subseteq A'$
 116. Prove or disprove:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

and

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

117. Prove or disprove: $(A \cap B) \cup C = A \cap (B \cup C)$
 118. Prove or disprove: $(A \cup B) - (A \cap B) = (A - B) \cup (B - A)$
Definition 41. Two sets A and B are disjoint if $A \cap B = \emptyset$.
 119. (*) Prove or disprove: $A \cap B$ and $A - B$ are disjoint.
 120. Prove or disprove: $A = (A \cap B) \cup (A - B)$.
 121. (*) Prove or disprove: $A - (A \cap B') = A \cap B$
 122. Prove or disprove: If A and B are sets such that $A \cup B = A \cap B$ then $A \cap B' = \emptyset$.
 123. (*) Prove or disprove: If A and B are sets such that $(A \cup B)' = A' \cup B'$ then $A = B$.
 124. (*) Prove or disprove: Let A , B and C be sets such that $A \cup B \neq A \cap C$. Then A is not a subset of C or B is not a subset of A .

Definition 42. A set of sets is called a family or collection of sets.

Example 43.

$$\mathcal{A} = \{\{1, 2, 3\}, \{3, 4, 5\}, \{3, 6\}, \{2, 3, 6, 7, 9, 10\}\}$$

is a family consisting of four sets.

125. Consider the family $\mathcal{A} = \{\{1, 2, 3\}, \{3, 4, 5\}, \{3, 6\}, \{2, 3, 6, 7, 9, 10\}\}$. Answer the following true/false questions, with reasons.
 (a) $\{3, 4, 5\} \in \mathcal{A}$
 (b) $3 \in \mathcal{A}$
 (c) $\{3, 4, 5\} \subseteq \mathcal{A}$

$$(d) \{\{3, 4, 5\}\} \subseteq \mathcal{A}$$

Definition 44. Let $\mathcal{A}$ be an family of sets. The union over $\mathcal{A}$ is

$$\begin{aligned} \bigcup_{A \in \mathcal{A}} A &= \{x \mid x \in A \text{ for some set } A \in \mathcal{A}\} \\ &= \{x \mid x \in A \text{ for at least one set } A \in \mathcal{A}\} \end{aligned}$$

Definition 45. Let $\mathcal{A}$ be an family of sets. The intersection over $\mathcal{A}$ is

$$\bigcap_{A \in \mathcal{A}} A = \{x \mid x \in A \text{ for every set } A \in \mathcal{A}\}$$

126. Consider the family $\mathcal{A} = \{\{1, 2, 3\}, \{3, 4, 5\}, \{3, 6\}, \{2, 3, 6, 7, 9, 10\}\}$. Answer the following, with explanations:

(a) Find $\bigcup_{A \in \mathcal{A}} A$

(b) Find $\bigcap_{A \in \mathcal{A}} A$

127. Let $A_n = \{1, 2, 3, \dots, n\}$ and let $\mathcal{A} = \{A_n \mid n \in \mathbb{N}\}$. Answer the following, with explanations:

(a) Find $\bigcup_{A_n \in \mathcal{A}} A_n$

(b) Find $\bigcap_{A_n \in \mathcal{A}} A_n$

128. Consider the family $\mathcal{A} = \{A_r = [r, \infty) \mid r \in \mathbb{R}\}$. Answer the following, with explanations:

(a) Find $\bigcup_{A_r \in \mathcal{A}} A_r$

(b) Find $\bigcap_{A_r \in \mathcal{A}} A_r$

129. For each natural number n , let $A_n = \left(0, \frac{1}{n}\right)$ and let $\mathcal{A} = \{A_n \mid n \in \mathbb{N}\}$. Answer the following, with explanations:

(a) Find $\bigcup_{A_n \in \mathcal{A}} A_n$

(b) Find $\bigcap_{A_n \in \mathcal{A}} A_n$

130. For each $n \in \mathbb{Z}$, let $C_n = [n, n+1)$ and let $\mathcal{C} = \{C_n \mid n \in \mathbb{Z}\}$. Answer the following, with explanations:

(a) Find $\bigcup_{C_n \in \mathcal{C}} C_n$

(b) Find $\bigcap_{C_n \in \mathcal{C}} C_n$

131. For each $n \in \mathbb{Z}$, let $A_n = (n, n+1)$ and let $\mathcal{A} = \{A_n \mid n \in \mathbb{Z}\}$. Answer the following, with explanations:

(a) Find $\bigcup_{A_n \in \mathcal{A}} A_n$

(b) Find $\bigcap_{A_n \in \mathcal{A}} A_n$

132. (*) Prove or disprove: For every set B in a family $\mathcal{A}$ of sets, $\bigcap_{A \in \mathcal{A}} A \subseteq B$

133. Prove or disprove: For every set B in a family $\mathcal{A}$ of sets, $B \subseteq \bigcup_{A \in \mathcal{A}} A$

134. (*) Prove or disprove: If the family $\mathcal{A}$ contains at least one set, then $\bigcap_{A \in \mathcal{A}} A \subseteq \bigcup_{A \in \mathcal{A}} A$.

Chapter 7

Relations

Definition 46. Let S be a set. Let $a, b \in S$. An ordered pair is a double (a, b) where a is the first term of the ordered pair and b is the second term of (a, b) . A relation R on S is a set of ordered pairs where the elements of the ordered pairs come from S .

Definition 47. The set of first terms is sometimes called the domain of the relation. The set of second terms is sometimes called the range of the relation.

Example 48. Let $S = \{2, 4, 6, 8, 10\}$. Then a relation on S could be defined by

$$R = \{(2, 2), (4, 4), (6, 6), (6, 8), (6, 10), (8, 6), (8, 8), (8, 10), (10, 6), (10, 8), (10, 10)\}.$$

This relation can be described as two numbers from S are related if they have the same number of divisors. It can also be written as $2R2$ meaning that 2 is related to 2 under the relation R .

135. Let $S = \{1, 2\}$. List all of the possible relations on S .

136. Let $A = \{1, 2, 3, 4, 5\}$. Write all elements of the following relations:

(a) $R = \{(a, b) \in A \times A \mid a \text{ divides } b\}$

(b) $E = \{(a, b) \in A \times A \mid a + b \text{ is even}\}$

(c) $U = \{(a, b) \in A \times A \mid a \neq b\}$

Definition 49. Let A and B be sets. Then $A \cup B$ is another set, and we can define a relation on $A \cup B$ called the Cartesian product of A and B , denoted $A \times B$, by $A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$. In other words, this relation is the set of all ordered pairs where the first term is an element of A and the second term is an element of B .

137. (*) Let $A = \{1, 2, 3\}$ and $B = \{2, 5\}$. List all elements of $A \times B$. What is the domain of this relation? What is the range of this relation?
138. Let $A = \{1\}$, $B = \{2\}$, and $C = \{3\}$. Show that $A \times (B \times C) \neq (A \times B) \times C$.
139. (*) Let A have n elements and B have m elements. Prove or disprove: $A \times B$ has mn elements.
140. Let A and B be non-empty sets. Prove that $A \times B = B \times A$ if and only if $A = B$.

Definition 50. Let S be a nonempty set and R a relation on S . Then

- (a) The relation R is reflexive on S if for every $x \in S$, $(x, x) \in R$.
 - (b) The relation R is symmetric on S if for every $x, y \in S$, $(y, x) \in R$ whenever $(x, y) \in R$.
 - (c) The relation R is transitive on S if for every $x, y, z \in S$, whenever $(x, y) \in R$ and $(y, z) \in R$ then $(x, z) \in R$.
 - (d) The relation R is antisymmetric on S if for every $x, y \in S$, whenever $(x, y) \in R$ and $(y, x) \in R$, then $x = y$.
141. Let $S = \{3, 5\}$. Is the relation $R = \{(3, 5)\}$ reflexive? symmetric? transitive? antisymmetric? Explain.
142. Let $S = \{1, 2, 3\}$. Find a relation on S that is reflexive and symmetric, but not transitive.
143. Let $S = \{1, 2, 3\}$. Find a relation on S that is reflexive and transitive, but not symmetric.
144. Let $S = \{1, 2, 3\}$. Find a relation on S that is symmetric and transitive, but not reflexive.

Definition 51. An equivalence relation on a set S is a relation that is reflexive, symmetric, and transitive.

145. Let $S = \{1, 2, 3\}$. Find a relation on S that is an equivalence relation.
146. Let $S = \{1, 2, 3\}$. Find a relation on S that is not reflexive, not symmetric, and not transitive.
147. Let $S = \mathbb{R}$. For any real numbers a and b , define $a \simeq b$ iff $a^2 = b^2$. Prove that $\simeq$ is an equivalence relation on $\mathbb{R}$. List all of the elements that are equivalent to -7 .
148. Let $S = \mathbb{R} \times \mathbb{R}$. Define $(a, b) \simeq (c, d)$ iff $a^2 + b^2 = c^2 + d^2$. Prove that $\simeq$ is an equivalence relation on $\mathbb{R} \times \mathbb{R}$. List all of the elements that are equivalent to $(0, 0)$.

149. Let $\mathcal{A} = \{\{1, 2\}, \{3, 4\}, \{5, 6, 7\}, \{8\}\}$.

- (a) List all elements of $\bigcup \mathcal{A}$
- (b) Let $R = \{(x, y) \in \bigcup \mathcal{A} \times \bigcup \mathcal{A} \mid x \text{ and } y \text{ belong to the same member of } \mathcal{A}\}$. Prove that R is an equivalence relation on $\bigcup \mathcal{A}$.
- (c) List all elements that are related to 5.

Definition 52. Let S be a non-empty set and R an equivalence relation on S . The set of all elements equivalent (i.e. related to) an element $a \in S$ are the equivalence class of a , denoted $[a]$.

150. For points (a, b) and (c, d) in $\mathbb{R}^2$, define $(a, b) \sim (c, d)$ iff $a^2 + b^2 = c^2 + d^2$.

- (a) Prove that $\sim$ is an equivalence relation on $\mathbb{R}^2$.
- (b) List all of the elements of $\mathbb{R}^2$ that are equivalent to $(0, 0)$.
- (c) List all of the elements of $\mathbb{R}^2$ that are equivalent to $(5, 11)$

151. Let the set $S = \{(x, y) \in \mathbb{R}^2 : y - x \text{ is an integer}\}$.

- (a) Prove that $\sim$ is an equivalence relation on $\mathbb{R}^2$.
- (b) List all of the elements of $\mathbb{R}^2$ that are equivalent to π .
- (c) List all of the elements of $\mathbb{R}^2$ that are equivalent to -17 .

152. For points (a, b) and (c, d) in $\mathbb{R}^2$, define $(a, b) \sim (c, d)$ iff $a^2 + b^2 = c^2 + d^2$.

- (a) Find $[(0, 0)]$
- (b) Find $[(5, 11)]$

153. Let the set $S = \{(x, y) \in \mathbb{R}^2 : y - x \text{ is an integer}\}$.

- (a) Find $[\pi]$.
- (b) Find $[-17]$.

Definition 53. A relation $\leq$ on a set A is a partial ordering if $\leq$ is reflexive, antisymmetric, and transitive.

154. Prove or disprove: For any set A , $\mathcal{P}(A)$ under $\subseteq$ is a partially ordered set.

155. Prove or disprove: For a set A , define an order $a \leq b$ iff $a = b$. Then A under $\leq$ is a partially ordered set.

Definition 54. A partially ordered set A with partial order $\leq$ is said to be totally ordered if given any two elements a and b in A , either $a \leq b$ or $b \leq a$.

156. Prove or disprove: For any set A , $\mathcal{P}(A)$ under $\subseteq$ is a totally ordered set.
157. (*) Prove or disprove: $\mathcal{P}(\mathbb{N})$ has an infinite totally ordered subset (under the order $\subseteq$).
158. Prove or disprove: For two integers a and b , define a relation by $a \sim b$ if $a \mid b$. Then this relation is **not** an equivalence relation.
159. Prove or disprove: For two integers a and b , define a relation by $a \sim b$ if $a \mid b$. Then this relation is **not** a total ordering.
160. Prove or disprove: For two integers a and b , define a relation by $a \sim b$ if $a \mid b$. Then this relation is a partial ordering.

Notes to the Instructor

¹ I am including the following, simple, truth-table as an example because it is one with which students should be familiar. They should be able to understand **why** the truth table looks as it does, both in terms of needing four rows and in terms of why the final statement is T or F in each instance. I make the students wrestle with the latter idea in problem 26 because it is often a source of confusion (i.e., most students do not understand why the conditional statement is true when the antecedent is false and the consequent is true), and having the students present this problem allows this conversation to happen (sometimes at length or repeatedly) in class, so that all students are given an opportunity to ask questions about that particular case. It also gives me as the instructor a chance to ask (or answer) many questions about this situation with all students present.

²The following problem is included here because it is a typical number theory problem that students will be expected to prove. I include it here as a demonstration that sometimes things that they **know** are true are difficult to prove. This problem is usually attempted (incorrectly) by several students before we get a correct solution, and so it remains open for quite a while. However, I want them to have this challenge early. You will notice that problem number 51 is a “generalization” of this problem. That one is included to make sure that students can understand and slightly alter the argument for problem 34. It is not uncommon for them to then have to prove that $\sqrt{p}$ for a prime p is irrational on the final exam.

³Parts (f) and (g) below are included to remind students about interval notation (which they should have seen in Calculus and is a prerequisite for this course). If your students have not seen that notation, you might include an appropriate definition here. The other purpose of those problems is to make sure that they understand the difference between $(6, 9]$ and $\{7, 8, 9\}$ since we have spent so much of the course focused on natural numbers and integers.